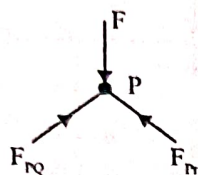


19. Ans: (b)

Sol: F.B.D of joint P is shown below,



For equilibrium of joint P,

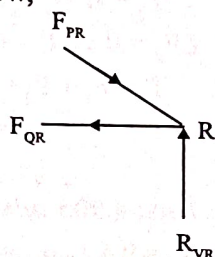
$$\sum F_x = 0 \Rightarrow F_{PQ} \cos 45 = F_{PR} \cos 30 \dots\dots (i)$$

$$\sum F_y = 0 \Rightarrow F_{PQ} \sin 45 + F_{PR} \sin 30 = F \dots\dots (ii)$$

From (i) and (ii)

$$F_{PR} = 0.732 F$$

F.B.D. of joint R is shown below,



For equilibrium of joint R,

$$F_{PR} \cos 30 = F_{QR}$$

$$\Rightarrow F_{QR} = 0.732 F \times \cos 30 = 0.63 F \text{ (Tensile)}$$

20. Ans: (b)

Sol: Torque about hinge, $\tau = P \times \frac{2L}{3} = \frac{2PL}{3}$

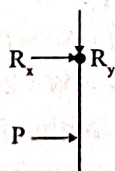
We know that, $\tau = I\alpha$

$$\frac{2PL}{3} = \frac{ML^2}{3} \times \alpha \Rightarrow \alpha = \frac{2P}{ML}$$

Tangential acceleration of centre of mass,

$$(a_t)_{cm} = \frac{L}{2} \times \alpha = \frac{L}{2} \times \frac{2P}{ML} = \frac{P}{M}$$

F.B.D of rod is shown below:

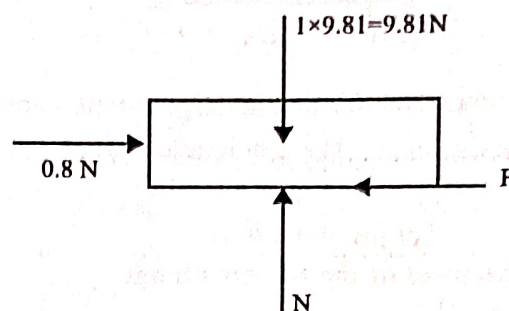


Net force along horizontal direction = Mass $\times$ Acceleration of centre of mass along horizontal direction

$$R_x + P = M \times \frac{P}{M} \Rightarrow R_x = 0$$

21. Ans: (b)

Sol:



$$\sum Y = 0 \Rightarrow N = 9.81 \text{ N}$$

$$F_s = \mu N = 0.1 \times 9.81 = 0.98 \text{ N}$$

The External force applied = $0.8 \text{ N} < F_s$

$\Rightarrow$ Frictional force = External applied force = 0.8 N

22. Ans: (a)

Sol: $\sum X = 0 \Rightarrow F = F_{AC} \cos 45 + F_{BC} \cos 60$

$$\sum Y = 0 \Rightarrow F_{AC} \sin 45 = F_{BC} \sin 60$$

$$F_{AC} = \frac{F_{BC} \sin 60}{\sin 45} = 1.224 F_{BC}$$

$$1 = 1.224 \times \cos 45 \times F_{BC} + F_{BC} \cos 60$$

$$1 = 0.865 F_{BC} + 0.5 F_{BC}$$

$$F_{BC} = \frac{1}{1.365} = 0.732 \text{ kN}$$

Vertical force at 'B' = $F_{BC} \sin 60$

$$= 0.732 \times \sin 60 = 0.634 \text{ kN}$$

23. Ans: (b)

$$\text{Sol: } \frac{F_{AC}}{\sin 120} = \frac{F}{\sin 105}$$

$$\Rightarrow F_{AC} = 0.895 F$$

F_{AC} = Maximum force

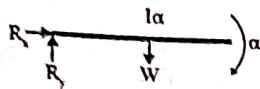
$$\frac{0.895 F}{\text{Area}} = \text{stress}$$

$$\Rightarrow \frac{0.895 F}{100} = 100$$

$$\Rightarrow F = 11.17 \text{ kN}$$

24. Ans: (b)

Sol: When the force 'F' is removed, then the F.B.D of rod is shown below



The rod accelerates and undergo rotation about the axis of the hinge. The acceleration ' α ' can be found by,

$$\alpha = \frac{\text{Torque about hinge}}{\text{Moment of inertia about hinge}}$$

$$\alpha = \frac{W \times \frac{L}{2}}{I}$$

$$I = \frac{W L^2}{g \cdot 3}$$

(moment of inertia of the rod about the hinge)

$$\therefore \alpha = \frac{W \cdot \frac{L}{2}}{\frac{W L^2}{g \cdot 3}} = \frac{3g}{2L} \left[\because I = \frac{mL^2}{3} \right]$$

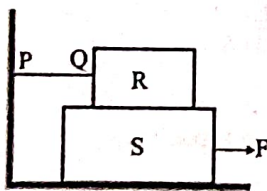
We know that, Force along any direction = Mass $\times$ Component of acceleration of centre of mass along that direction

$$\therefore W - R_y = m \times \frac{L}{2} \alpha = \frac{3mg}{4}$$

$$\therefore R_y = \frac{mg}{4} = \frac{W}{4}$$

25. Ans: (d)

Sol:



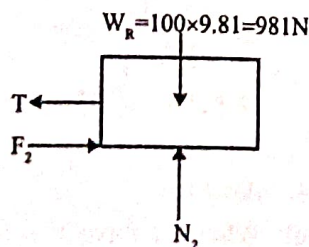
FBD of block R :

$$\Sigma F_y = 0$$

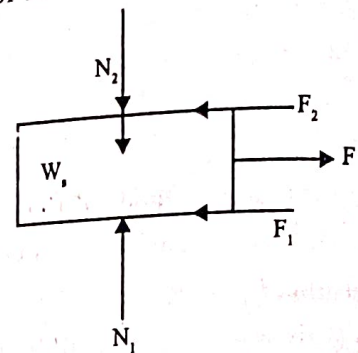
$$N_2 = W_R = 981 \text{ N}$$

$$F_2 = \mu N_2$$

$$= 0.4 \times 981 = 392.4 \text{ N}$$



FBD of block S:



$$W_s = 150 \times 9.81 = 1471.5 \text{ N}$$

$$\Sigma F_y = 0$$

$$N_1 = N_2 + W_s = 981 + 1471.5 = 2452.5 \text{ N}$$

$$F_1 = \mu N_1 = 0.4 \times 2452.5 = 981 \text{ N}$$

$$\Sigma F_x = 0 \Rightarrow F - F_1 - F_2 = 0$$

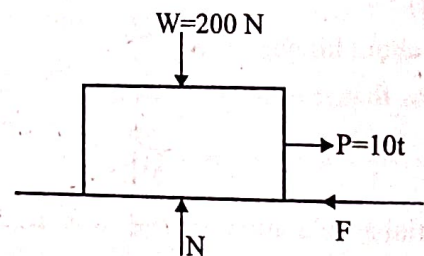
$$F = F_1 + F_2 = 981 + 392.4$$

$$= 1373.4 \text{ N} = 1.3734 \text{ kN}$$

26. Ans: 4.905 m/s

Sol: $\mu_s = 0.4$; $\mu_k = 0.2$

FBD of the block



$$F_s = \mu_s N$$

$$F_k = \mu_k N$$

$$\Sigma F_y = 0$$

$$N - W = 0$$

$$N = W = 200 \text{ N}$$

Limiting friction or static friction

$$F_s = 0.4 \times 200 = 80 \text{ N}$$

Kinetic Friction

$$F_k = 0.2 \times 200 = 40 \text{ N}$$

The block starts moving only when the force, P exceeds static friction, F_s

Thus, under static equilibrium

$$\Rightarrow \Sigma F_x = 0$$

$$\Rightarrow P - F_s = 0 \Rightarrow 10t = 80$$

$$t = \frac{80}{10} = 8 \text{ sec}$$

∴ The block starts moving only when $t > 8$ seconds

During 8 seconds to 10 seconds of time:

According to Newton's second law of motion

Force = mass × acceleration

$$(P - F_k) = m \times \frac{dv}{dt} \Rightarrow (10t - 40) = \frac{200}{9.81} \times \frac{dv}{dt}$$

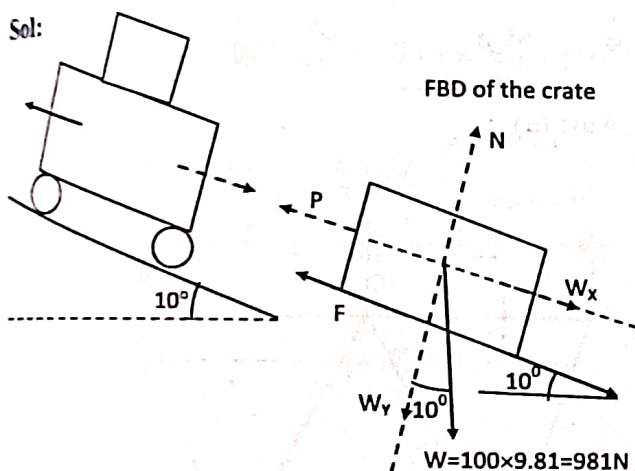
$$\int_8^{10} (10t - 40) dt = \frac{200}{9.81} \int_0^v dv$$

$$[5t^2 - 40t]_8^{10} = 20.387 \times V \Rightarrow (180 - 80) = 20.387 \times V$$

$$\text{Velocity (V)} = 4.905 \text{ m/s}$$

27. Ans: 1.198 m/s²

Sol:



w.r.t. FBD of the crate:

$$W_x = W \sin 10^\circ = 981 \times \sin 10^\circ = 170.34 \text{ N}$$

$$W_y = W \cos 10^\circ = 981 \times \cos 10^\circ = 966.09 \text{ N}$$

$$\Sigma F_y = 0 \Rightarrow N - W_y = 0$$

$$N = W_y = 966.09 \text{ N};$$

$$F = \mu N = 0.3 \times 966.09 = 289.828 \text{ N}$$

$$\Sigma F_x = 0 \Rightarrow P = W_x - F$$

$$\Rightarrow P + 289.828 - 170.34 = 0$$

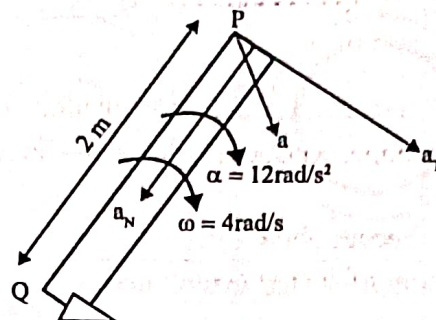
$$P = 119.488 \text{ N}$$

$$P = ma = 119.488 \text{ N}$$

$$\Rightarrow a = \frac{119.488}{100} = 1.198 \text{ m/s}^2$$

28. Ans: 40 m/s²

Sol:



Tangential acceleration

$$a_T = r \alpha = 2 \times 12 = 24 \text{ m/s}^2$$

$$\text{Normal acceleration, } a_N = r \omega^2 = 2 \times 4^2 = 32 \text{ m/s}^2$$

The resultant acceleration

$$a = \sqrt{a_T^2 + a_N^2} = \sqrt{24^2 + 32^2} = 40 \text{ m/s}^2$$

29. Ans: 57.67 m

Sol:

$$W_x = W \sin 45^\circ$$

$$= 98.1 \times \sin 45^\circ$$

$$= 69.367 \text{ N}$$

$$W_y = W \cos 45^\circ$$

$$= 69.367 \text{ N}$$

$$\Sigma F_y = 0$$

$$N - W_y = 0$$

$$N = W_y$$

$$= 69.367 \text{ N}$$

$$F = \mu_k N$$

$$= 0.5 \times 69.367$$

$$= 34.683 \text{ N}$$

$$\Sigma F_x = 0 \text{ (Dynamic Equilibrium)}$$

– D' Alembert's Principle)

$$W_x - F - ma = 0$$

$$69.367 - 34.683 - 10 \times a = 0$$

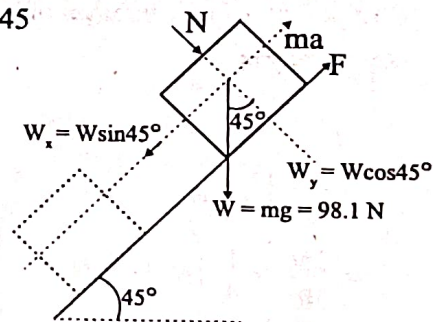
$$a = 3.468 \text{ m/s}^2$$

$$V = 20 \text{ m/s}^2$$

$$u = 0;$$

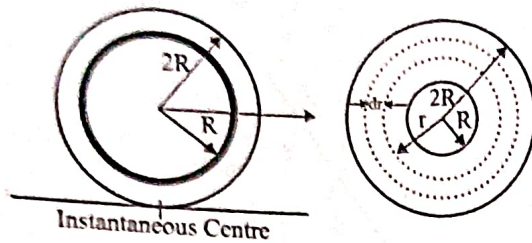
$$V^2 = 2as$$

$$s = \frac{V^2}{2a} = \frac{20^2}{2 \times 3.468} = 57.67 \text{ m}$$



30. Ans: (c)

Sol:



Mass per unit surface area of disc, $= \frac{M}{\pi[(2R)^2 - R^2]}$

$$= \frac{M}{\pi(4R^2 - R^2)} = \frac{M}{3\pi R^2}$$

Mass of elementary ring,

$$m = \frac{M}{3\pi R^2} \times 2\pi r dr = \frac{2M}{3R^2} r dr$$

Moment of inertia of that elementary ring is given by,

$$dI = \frac{2M}{3R^2} r dr \times r^2$$

As such elementary rings vary from $r = R$ to $r = 2R$, the total moment of inertia of whole disc is given by

$$I = \int_R^{2R} \frac{2M}{3R^2} r^3 dr$$

$$\begin{aligned} I &= \frac{2M}{3R^2} \left[\frac{r^4}{4} \right]_R^{2R} \\ &= \frac{2M}{3R^2} \times \frac{1}{4} [16R^4 - R^4] \\ &= \frac{2M}{3R^2} \times \frac{1}{4} \times 15R^4 = \frac{5MR^2}{2} \end{aligned}$$

$$\text{Total kinetic energy} = \frac{1}{2} m V^2 + \frac{1}{2} I \omega^2$$

$$V = r\omega = 2R\omega$$

$$\omega = \frac{V}{2R} \text{ rad/s}$$

Total Kinetic Energy

$$\begin{aligned} &= \frac{1}{2} m V^2 + \frac{1}{2} \times \frac{5mR^2}{2} \times \left(\frac{V}{2R} \right)^2 \\ &= \frac{1}{2} m V^2 + \frac{5mV^2}{16} = \frac{13mV^2}{16} \end{aligned}$$

31. Ans: 10 N - m

Sol: $m = 1000 \text{ kg}$,

$$\omega_0 = 10 \text{ rad/sec},$$

$$\omega = \omega_0 + \alpha t$$

$$0 = 10 + \alpha \times 10$$

$$\alpha = -1 \text{ rad/sec}^2$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$= 10 \times 10 + \frac{1}{2} (-1) \times 10^2 = 100 - 50$$

$$= 100 - 50$$

$$\theta = 50 \text{ rad}$$

$$\text{Work done} = \frac{1}{2} m (V_2^2 - V_1^2) = \frac{1}{2} m r^2 (\omega^2 - \omega_0^2)$$

$$T\theta = \frac{1}{2} \times 1000 \times 0.2^2 \times (-100)$$

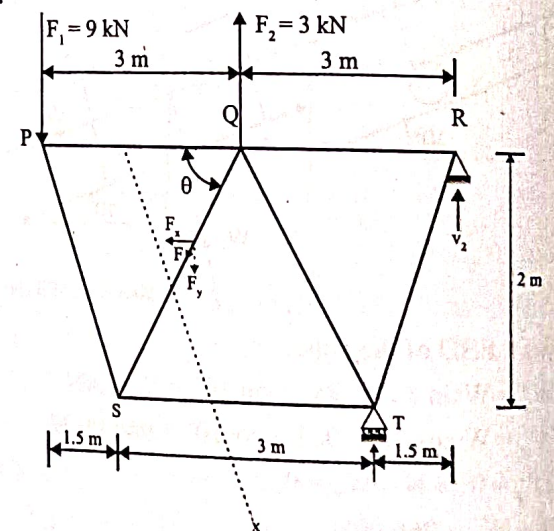
$$T \times 50 = 2000$$

$$T = 40 \text{ N-m (anti-clock wise)}$$

$$\text{Torque per wheel} = \frac{40}{4} = 10 \text{ N-m}$$

32. Ans: (a)

Sol:



$$\Sigma F_y = 0$$

$$V_1 + V_2 - 9 + 3 = 0$$

$$\Sigma M_R = 0$$

$$\Rightarrow V_1 \times 1.5 + 3 \times 3 - 9 \times 6 = 0$$

$$\Rightarrow V_1 = 30 \text{ kN } (\uparrow)$$

$$\Rightarrow V_2 = -30 + 9 - 3 = -24 \text{ kN } (\downarrow)$$

$$\theta = \tan^{-1}\left(\frac{2.0}{1.5}\right) = 53.13^\circ$$

$\Sigma F_y = 0$ (w.r.t. RHS of the section x-x)

$$V_1 + F_2 - V_2 - F_y = 0 \quad F_2$$

$$\Rightarrow F \sin 53.13 = 30 + 3 - 24$$

$$F = 11.25 \text{ kN (Tension)}$$

$\therefore$ Force in member, QS = 11.25 kN (Tension)

(or)

$$F_1 = F_{PS} \sin(53.13)$$

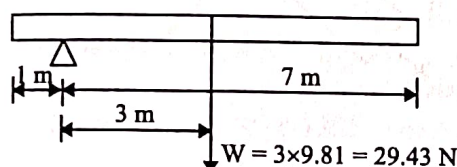
$$F_{PS} = 11.25 \text{ compression}$$

$$-F_{PS} = F_{QS}$$

$$\therefore F_{QS} = 11.251 \text{ (Tension)}$$

33. Ans: 2.053 rad/s²

Sol:



$$M = I\alpha$$

$$M = 29.43 \times 3 = 88.29 \text{ N-m}$$

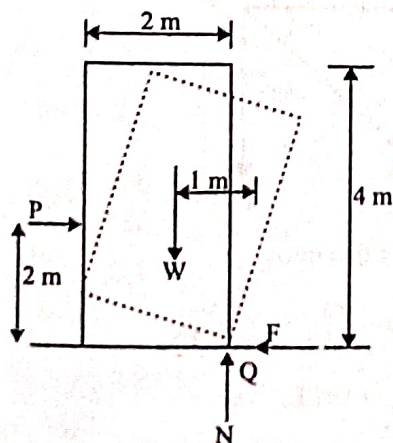
$$I = I_o + md^2 = \frac{ml^2}{12} + md^2 = \frac{3 \times 8^2}{12} + 3 \times 3^2$$

$$= 16 + 27 = 43 \text{ kg-m}^2$$

$$\alpha = \frac{M}{I} = \frac{88.29}{43} = 2.053 \text{ rad/s}^2$$

34. Ans: (a)

Sol:



$$\Sigma M_Q = 0$$

$$\Rightarrow P \times 2 - W \times 1 = 0$$

$$P = \frac{100 \times 9.81}{2} = 490.5 \text{ N}$$

$$\Sigma F_y = 0$$

$$\Rightarrow N = W = 981 \text{ N}$$

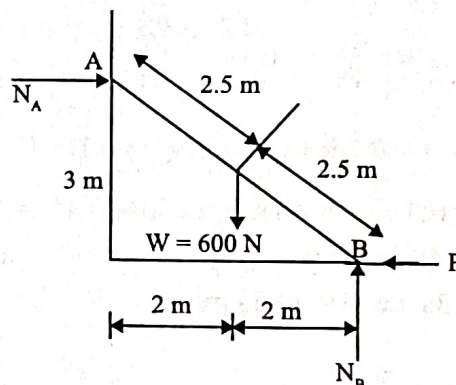
$$F = \mu N = P$$

$$490.5 = \mu \times 981$$

$$\mu = \frac{490.5}{981} = 0.5$$

35. Ans : 399 to 401 N

Sol:



$$\Sigma F_y = 0$$

$$N_B - W = 0 \Rightarrow N_B = 600 \text{ N}$$

$$\Sigma M_A = 0$$

$$\therefore P \times 3 + W \times 2 - N_B \times 4 = 0$$

$$P = \frac{4N_B - 2W}{3}$$

$$P = \frac{4 \times 600 - 2 \times 600}{3} = 400 \text{ N}$$

36. Ans: (c)

Sol: PQ = QR = 4m

Take moments about Q

$$100 \cos 60^\circ \times PQ = V_R \times 4$$

$$\Rightarrow V_R = 50 \text{ kN}$$

$$\Sigma F_y = 0$$

$$V_R = F_{PR} \sin 45^\circ$$

$$F_{PR} = \frac{50}{\sin 45^\circ} = 70.71 \text{ kN}$$

37. Ans: 0.64

Sol: Let, V_1 = Velocity with which ball hits the ground

$$V_1 = \sqrt{2gh_1}$$

$$V_1 = \sqrt{2 \times 9.81 \times 1} = 4.427 \text{ m/sec}$$

$$\begin{aligned} \text{Velocity with which ball bounces up} \\ = 4.427 - 20\% \text{ of } 4.427 \\ = 3.54 \text{ m/sec} \end{aligned}$$

$$h_2 = \frac{V^2}{2g} = \frac{3.54^2}{2 \times 9.81} = 0.64 \text{ m}$$

38. Ans: 29.5 to 29.7

Sol: $v_o = 40 \text{ m/s}$;

$$a = -0.1v \Rightarrow \frac{dv}{dt} = -0.1v$$

$$\int \frac{dv}{v} = \int -0.1 dt + C_1 \Rightarrow \ln v = -0.1t + C_1$$

$$\text{When } t = 0 \Rightarrow v = 40 \text{ m/s} \Rightarrow \ln 40 = C_1 = 3.68$$

$$\ln v = 3.68 - 0.1t$$

$$\text{At, } t = 3 \text{ s} \Rightarrow V = 29.63 \text{ m/s}$$

39. Ans: (c)

Sol: Take moments about point P

$$V_o \times 3 = 30 \times 2$$

$$V_o = 20 \text{ kN}$$

At point Q,

$$\Sigma F_y = 0$$

$$F_{RQ} = V_o$$

$$F_{RQ} = 20 \text{ kN}$$

At point R,

$$\Sigma F_y = 0$$

$$F_{RQ} = F_{RT} \sin 45$$

$$F_{RT} = \frac{20}{\sin 45}$$

$$\Sigma F_x = 0$$

$$F_{RQ} = F_{RT} \cos 45$$

$$= \frac{20}{\sin 45} \times \cos 45$$

$$F_{RT} = 20 \text{ kN}$$

40. Ans: 251 to 252

$$\text{Sol: } \frac{\theta}{2\pi} = \frac{S}{P}$$

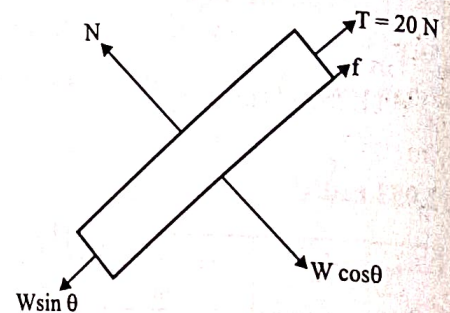
$$\frac{\dot{\theta}}{2\pi} = \frac{\dot{S}}{P}, \quad \dot{\theta} = \omega, \quad \dot{S} = V$$

$$\omega = \frac{V \times 2\pi}{P} = \frac{20 \times 2\pi}{0.5}$$

$$\Rightarrow \omega = 80\pi \text{ rad/sec} = 251.32 \text{ rad/sec}$$

41. Ans: 5 (range 4.95 to 5.05)

Sol:



$$\Sigma F_y = 0$$

$$\Rightarrow N = W \cos \theta = 0.8 W$$

$$f = \mu N = 0.2 W$$

$$\Sigma F_x = 0$$

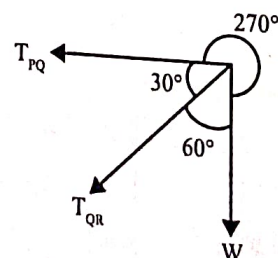
$$\Rightarrow 0.6 W = 20 + 0.2 W$$

$$\Rightarrow W = 50 \text{ N}$$

$$\Rightarrow m = 5 \text{ kg}$$

42. Ans: (b)

Sol:



By Lami's theorem

$$\frac{W}{\sin 30} = \frac{T_{PQ}}{\sin 60} = \frac{T_{QR}}{\sin 270}$$

$$\Rightarrow T_{PQ} = \sqrt{3} W (T)$$

$$\Rightarrow T_{QR} = -2 W = 2 W (C)$$

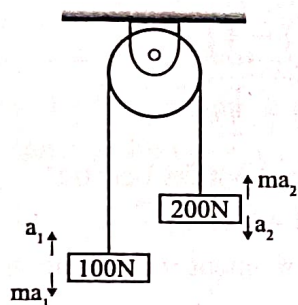
43. Ans: (b & d)

Sol: The rate of change of the angular momentum of the particles about 'G' is

$$\begin{aligned} &= \frac{d}{dt} (\Sigma \mathbf{m} \times \rho \times \bar{\mathbf{v}}_i) \\ &= \Sigma \left(m \frac{d\rho}{dt} \times \mathbf{v}_i + m\rho \times \frac{d\bar{\mathbf{v}}_i}{dt} \right) \\ &= -\Sigma \mathbf{v}_i \times \frac{d\rho}{dt} + \Sigma m\rho \times \frac{d\bar{\mathbf{v}}_i}{dt} \\ &= -\mathbf{v}_i \times \frac{d}{dt} (\Sigma m\rho) + \Sigma m\rho \times \bar{\mathbf{r}}_i \\ &= \bar{\mathbf{0}} + \Sigma m\rho \times \bar{\mathbf{r}}_i = \Sigma m_i \rho \times \bar{\mathbf{r}}_i \\ &= \Sigma m_i \rho \times (\bar{\mathbf{r}}_G + \bar{\rho}_i) \\ &= \Sigma m_i \rho \times \bar{\mathbf{r}}_G + \Sigma m_i \rho \times \bar{\rho}_i \\ &= -\bar{\mathbf{r}}_G \times \Sigma m_i \rho + \Sigma m_i \rho \times \bar{\rho}_i \\ &= \bar{\mathbf{0}} + \Sigma m_i \rho \times \bar{\rho}_i = \Sigma m_i \rho \times \bar{\rho}_i \end{aligned}$$

44. Ans: 133.33 (Range 130 to 135)

Sol:



$$m_1 a_1 = T - 100$$

$$\frac{100}{g} a_1 = T - 100 \quad \text{----- (1)}$$

$$m_2 a_2 = 200 - T$$

$$\frac{200}{g} a_2 = 200 - T \quad \text{----- (2)}$$

$$\frac{200}{g} a = 200 - \frac{100}{g} a - 100$$

$$300 a = 100 g$$

$$\Rightarrow a = \frac{g}{3} = \frac{10}{3}$$

$$\therefore T = 100 a_1 + 100 \quad [\because a_1 = a_2]$$

$$T = \frac{100}{g} \times \frac{g}{3} + 100 = 133.33 \text{ N}$$

45. Ans: 20 (Range: 19.9 to 20.1)

$$\text{Sol: } \frac{1}{2} I \omega^2 - 0 = mgh$$

$$\frac{1}{2} \times \frac{3}{2} m r^2 \times \omega^2 = mgh$$

$$\frac{3}{4} m v^2 = mgh$$

$$v^2 = \frac{4}{3} gh$$

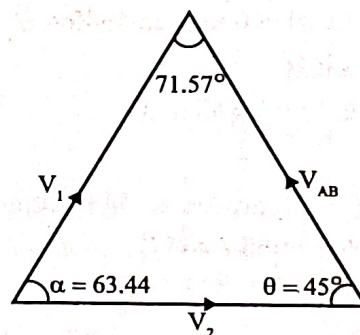
$$v = \sqrt{\frac{4}{3} gh} = \sqrt{\frac{4}{3} \times 10 \times 30} = 20 \text{ m/sec}$$

46. Ans: 3 (range 2.9 to 3.1)

$$\text{Sol: } \mathbf{V}_1 = \mathbf{i} + 2\mathbf{j}$$

$$|\mathbf{V}_1| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

By drawing velocity diagram



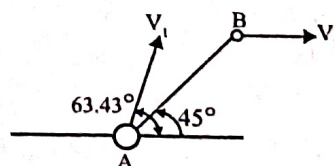
$$\alpha = \tan^{-1} \left(\frac{2}{1} \right) = 63.44$$

By applying sine rule

$$\frac{\sqrt{5}}{\sin 45^\circ} = \frac{V_2}{\sin 71.57^\circ}$$

$$\therefore V_2 = 3 \text{ m/sec}$$

Alternate solution:



As the rod is rigid the velocity of all the points on rod along the length must be same.

$$V_1 = \sqrt{2^2 + 1^2} = \sqrt{5} \text{ m/s}$$

$$\alpha = \tan^{-1}\left(\frac{2}{1}\right) = 63.44^\circ$$

$$V_1 \cos(63.43 - 45) = V_2 \cos 45$$

$$\Rightarrow V_2 = 3 \text{ m/s}$$

47. Ans: (680)

Sol: $V_w = 800 \text{ m/s}$,

$$m_w = 3.4 \text{ kg/min},$$

$$m_a = 0.6 \text{ kg/min}$$

$$\dot{m}_a v_a + \dot{m}_w v_w = (\dot{m}_a + \dot{m}_w) V$$

$$\frac{0.6}{60} \times 0 + \frac{3.4}{60} \times 800 = \left(\frac{0.6 + 3.4}{60}\right) \times V$$

$$45.33 = \frac{1}{15} \times V$$

$$V = 45.33 \times 15 = 679.95 \approx 680 \text{ m/sec}$$

48. Ans: (a)

Sol: $V = R\omega$

$$I_A = mR^2, \quad I_B = \frac{mR^2}{2}$$

$$\Rightarrow \frac{1}{2} m V_A^2 + \frac{1}{2} I_A \omega_A^2 = \frac{1}{2} m V_B^2 + \frac{1}{2} I_B \omega_B^2$$

$$\Rightarrow \frac{1}{2} m \times V_A^2 + \frac{1}{2} m R^2 \times \left(\frac{V_A}{R}\right)^2 = \frac{1}{2} m V_B^2 + \frac{1}{2} \times \frac{m R^2}{2} \times \left(\frac{V_B}{R}\right)^2$$

$$\Rightarrow \frac{1}{2} m V_A^2 + \frac{1}{2} m V_A^2 = \frac{1}{2} m V_B^2 + \frac{1}{4} m V_B^2$$

$$\Rightarrow m V_A^2 = \frac{3m V_B^2}{4}$$

$$\therefore \frac{V_A}{V_B} = \sqrt{\frac{3}{4}}$$

49. Ans: 7.5

Sol: Given:

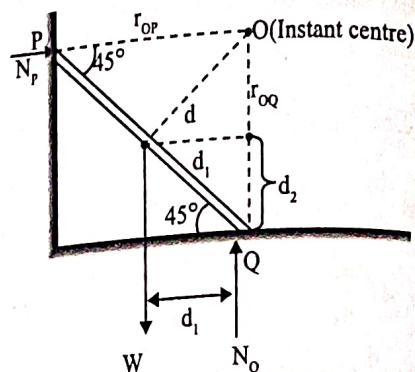
$$L = \sqrt{2} \text{ m},$$

$$m = 10 \text{ kg},$$

$$g = 10 \text{ m/s}^2,$$

$$I_{CG} = \frac{mL^2}{12}$$

Angular acceleration, $\alpha = ?$



The rod PQ is in general plane motion while falling down.

$$d_1 = \frac{\sqrt{2}}{2} \times \cos 45^\circ$$

$$d_1 = \frac{1}{2} m = d_2$$

$$r_{OQ} = \sqrt{2} \times \sin 45^\circ = 1 \text{ m} = r_{OP}$$

$$d^2 = d_1^2 + (r_{OQ} - d_2)^2$$

$$d^2 = \left(\frac{1}{2}\right)^2 + \left(1 - \frac{1}{2}\right)^2$$

$$d^2 = 2 \times \left(\frac{1}{2}\right)^2 = \frac{1}{2} \text{ m}^2$$

Take moment about the I centre

$$\text{Moment} = M = I_O \alpha$$

$$I_O = \text{Mass moment of inertia with respect to instantaneous centre O} = I_{CG} + md^2 = \frac{m\ell^2}{12} + md^2$$

$$I_O = \frac{m}{12} (\sqrt{2})^2 + m \times \frac{1}{2} = \frac{m}{6} + \frac{m}{2} = \frac{4m}{6} = \frac{2m}{3}$$

α = Angular acceleration

$$M = \text{moment with respect to O} = W \times d_1 = m \times g \times d_1$$

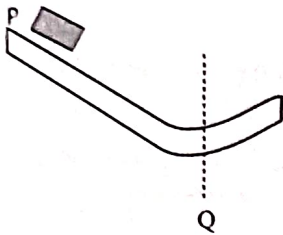
$$m \times g \times d_1 = \frac{2m}{3} \times \alpha$$

$$\Rightarrow \alpha = \frac{3}{2} \times 10 \times \frac{1}{2}$$

$$= 7.5 \text{ rad/sec}^2$$

50. Ans: 420

Sol:



Given:

$$m = 2 \text{ kg},$$

$$r = 2 \text{ m},$$

$$g = 10 \text{ m/s}^2,$$

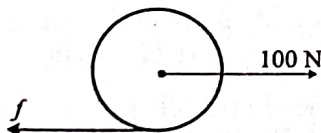
$$V = 20 \text{ m/s},$$

$$\begin{aligned} \text{Total normal force} &= W + \frac{mv^2}{r} \\ &= mg + \frac{mv^2}{r} \\ &= (2 \times 10) + \frac{(2 \times 20^2)}{2} \\ &= 20 + 400 = 420 \text{ N} \end{aligned}$$

51. Ans: 6.66

Sol: $F = 100 \text{ N}$, $r = 1 \text{ m}$, $m = 10 \text{ kg}$

The body is having the combined effect of translation and rotation



(i) Translation :

$$\Sigma F_x = \text{max}$$

$$(100 - F) = 10 \times a \text{ ---- (1)}$$

(ii) Rotation :

$$T = I\alpha$$

$$fr = I\alpha$$

$$fr = \frac{mr^2}{2} \times \alpha$$

$$f = \frac{ma}{2} \text{ ---- (2)}$$

$$\left(100 - \frac{mr}{2} \times a\right) = 10 \times a$$

$$\left(100 - \frac{10 \times 1}{2} \times a\right) = 10 \times a$$

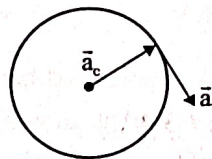
$$100 = 10a + \frac{10a}{2}$$

$$100 = 15a$$

$$\Rightarrow a = 6.66 \text{ m/s}^2$$

52. Ans: (a)

Sol:



$$R = 1 \text{ m}, \alpha = 1 \text{ rad/s}^2, \omega = 1 \text{ rad/s}$$

$$\vec{a}_{QT} = \vec{a}_c + \vec{a}_t$$

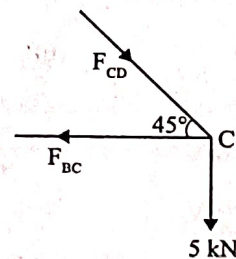
$$a_c = R\omega^2 \text{ and } a_t = R\alpha$$

$$\begin{aligned} \therefore \text{Total acceleration} &= \sqrt{1^2 + 1^2} \\ &= \sqrt{2} = 1.414 \text{ m/s}^2 \end{aligned}$$

53. Ans: 5

Sol: BD is a zero force member

F.B.D of joint C is shown below,



For equilibrium of joint C,

$$\Sigma F_y = 0, F_{CD} \cos 45 = 5$$

$$\therefore F_{CD} = \frac{5}{\cos 45}$$

$$\Sigma F_x = 0,$$

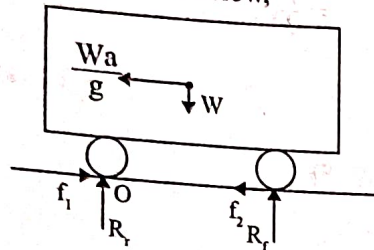
$$F_{BC} = F_{CD} \cos 45$$

$$= \frac{5}{\cos 45} \times \cos 45 = 5 \text{ kN}$$

54. Ans: (b)

Sol: We analyse this problem in the frame of reference of car.

F.B.D of car is shown below,



As our frame of reference is accelerated hence, we have to apply a pseudo force 'ma' as shown above, where, f_1 and f_2 are friction forces on rear and front wheels respectively.

For vertical equilibrium,

$$R_r + R_f = W \quad \text{---(i)}$$

$$\sum M_o = 0,$$

$$W \times \frac{\ell}{2} - \frac{W a}{g} \times h - R_f \times \ell = 0 \quad \text{---(ii)}$$

From (i) & (ii)

$$R_f = \frac{W}{2} - \frac{W}{g} \left(\frac{h}{\ell} \right) a$$

$$R_r = \frac{W}{2} + \frac{W}{g} \left(\frac{h}{\ell} \right) a$$

55. Ans: (d)

Sol: Let V_1 is the speed of 3 kg mass after collision
 V_2 is the speed of m kg mass after collision

$$e = 1 = \frac{V_2 - V_1}{4}$$

$$\Rightarrow V_2 - V_1 = 4 \quad \text{---(1)}$$

By linear momentum conservation

$$3 \times 4 = 3V_1 + mV_2 \quad \text{---(2)}$$

$$\frac{1}{2} \times 3 \times V_1^2 = 6$$

$$\Rightarrow V_1 = \pm 2 \quad \text{---(3)}$$

By using (1), (2) & (3) we get

$$m = 1 \text{ kg (or) } 9 \text{ kg}$$

56. Ans: (b)

Sol: By using conservation of mechanical energy

$$E_i = E_f$$

$$\frac{1}{2} m v_i^2 = \frac{1}{2} m v_f^2 + \frac{1}{2} k \Delta x^2$$

$$\frac{1}{2} \times 2 \times 1.5^2 = \frac{1}{2} \times 2 \times v_f^2 + \frac{1}{2} \times 5 \times 0.4^2$$

$$v_f = 1.36 \text{ m/s}$$

57. Ans: 1

Sol: Let, V is the velocity with which the two mass moves after sticking,

$\therefore$ By using Linear momentum conservation

$$m_1 V_o = (m_1 + m_2) V$$

$$10 \times 2 = (10 + 30) V$$

$$V = \frac{1}{2} \text{ m/s}$$

At the instant of maximum compression in spring the two stuck mass comes to rest.

$\therefore$ By using conservation of mechanical energy.

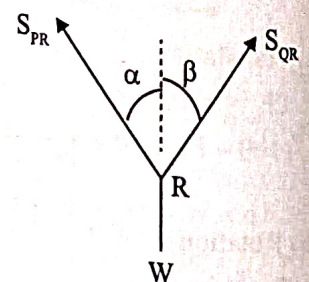
$$\frac{1}{2} (m_1 + m_2) V^2 = \frac{1}{2} K x^2$$

$$\frac{1}{2} \times 40 \times \left(\frac{1}{2} \right)^2 = \frac{1}{2} \times 10^5 \times x^2$$

$$\Rightarrow x = 0.01 \text{ m} = 1 \text{ cm}$$

58. Ans: (c)

Sol: FBD at joint R :



Using Lami's theorem

$$\frac{W}{\sin(\alpha + \beta)} = \frac{S_{QR}}{\sin(180 - \alpha)}$$

$$\therefore S_{QR} = \frac{W \times \sin \alpha}{\sin(\alpha + \beta)}$$

59. Ans: (c)

Sol: Calculation of Kinetic Energy (KE) of block:

$$\text{K.E of block} = \frac{1}{2} \times M \times V^2 = \frac{1}{2} \times M \times \dot{x}^2 \dots\dots(i)$$

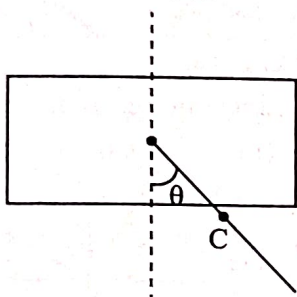
Calculation of KE of rod:

Kinetic energy of a body (mass m) in general plane motion is given by,

$$\text{K.E} = \frac{1}{2} \times m \times V_{cm}^2 + \frac{1}{2} \times I_{cm} \times \omega^2$$

where, V_{cm} = Speed of centre of mass of body.

I_{cm} = Mass moment of inertia of body about centre of mass.



Position of centre of mass (c),

$$x \text{ co-ordinate} = x + \frac{b}{2} \sin \theta$$

Velocity of centre of mass along x direction

$$V_x = \frac{dx}{dt} = \dot{x} + \frac{b}{2} \cos \theta \times \dot{\theta}$$

$$y \text{ co-ordinate} = \frac{-b}{2} \cos \theta$$

Velocity of centre of mass along y direction

$$V_y = \frac{dy}{dt} = \frac{b}{2} \sin \theta \times \dot{\theta} \quad \left[\dot{\theta} = \frac{d\theta}{dt} = \omega \right]$$

$$\text{K.E of rod} = \frac{1}{2} \times m \times [V_x^2 + V_y^2] + \frac{1}{2} \times \frac{mb^2}{12} \times \dot{\theta}^2$$

$$= \frac{1}{2} \times m \left[\left(\dot{x} + \frac{b}{2} \dot{\theta} \cos \theta \right)^2 + \left(\frac{b}{2} \sin \theta \times \dot{\theta} \right)^2 \right] + \frac{1}{2} \times \frac{mb^2}{12} \times \dot{\theta}^2$$

$$= \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m b \dot{\theta} \dot{x} \cos \theta + \frac{1}{6} m b^2 \dot{\theta}^2 \dots\dots(ii)$$

$$\text{P.E of spring} = \frac{1}{2} kx^2 \dots\dots(iii)$$

$$\text{P.E of rod} = m \times g \times h$$

[where, h = height of center of mass from a reference level]

$$\text{P.E of rod} = -m \times g \times \frac{b}{2} \cos \theta$$

[Reference level is taken as floor]

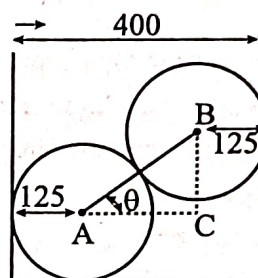
$$\text{Lagrangian} = \text{KE} - \text{PE}$$

$$= \frac{1}{2} (M + m) \dot{x}^2 + \frac{1}{2} m b \dot{\theta} \dot{x} \cos \theta + \frac{1}{6} m b^2 \dot{\theta}^2$$

$$- \frac{1}{2} kx^2 + mg \frac{b}{2} \cos \theta$$

60. Ans: (125)

Sol:



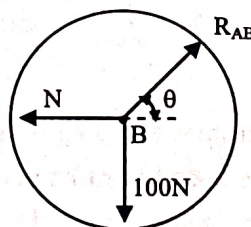
$$l(AC) = 400 - 125 - 125 = 150$$

$$l(AB) = 125 + 125 = 250$$

$$\therefore \cos \theta = \frac{150}{250} = \frac{3}{5}$$

$$\therefore \sin \theta = \frac{4}{5}$$

FBD for Upper Cylinder:



$$\Sigma F_y = 0$$

$$\therefore R_{AB} \times \sin \theta = 100$$

$$\therefore R_{AB} \times \frac{4}{5} = 100$$

$$\therefore R_{AB} = 125 \text{ N}$$

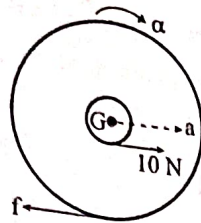
61. Ans: (1.4)

Sol: Let,

f = friction force,

a = acceleration of centre of mass,

α = angular acceleration of wheel



$$I_G = 2 \times (M \times r^2) = 2 \times 1 \times 0.4^2 = 0.32 \text{ kg-m}^2$$

$$\Sigma F = m \cdot a$$

$$\Rightarrow 10 - f = 3.5 \times a$$

$$\Sigma T = I \cdot \alpha$$

$$\Rightarrow -10 \times 0.1 + f \times 0.4 = 0.32 \times \frac{a}{0.4} \quad \dots\dots(ii)$$

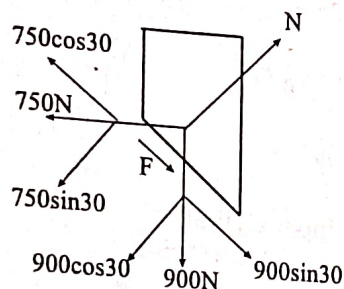
[As there is no slip $\therefore a = r\alpha$]

Solving (i) and (ii),

$$\therefore a = 1.36 \text{ m/s}^2$$

62. Ans: (0.17)

Sol: FBD of block



From FBD,

$$750 \cos 30 > 900 \sin 30$$

$\therefore$ Block will try to slide upward

$\therefore$ Frictional force on block will be downward

$$N = 750 \sin 30 + 900 \cos 30 = 1154.4 \text{ (Newton)}$$

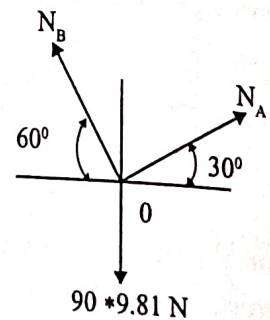
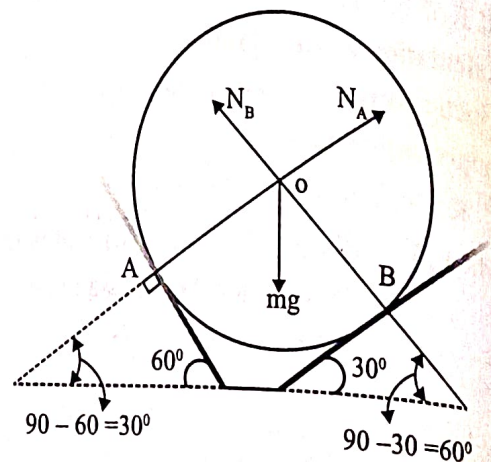
$$\& 750 \cos 30 - 900 \sin 30 = f = \mu \times N$$

$$= \mu \times 1154.4$$

$$\therefore \mu = 0.1728$$

63. Ans: (147.15)

Sol: FBD at centre of sphere:



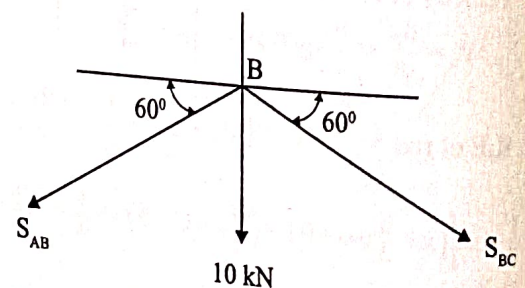
Using Lami's theorem,

$$\frac{30 \times 9.81}{\sin 90} = \frac{N_A}{\sin 150}$$

$$\therefore N_A = 147.15 \text{ N}$$

64. Ans: (-5773.5)

Sol: FBD at joint B:



Using Lami's theorem:

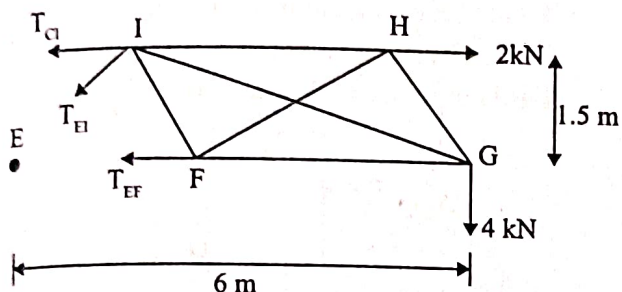
$$\frac{10}{\sin 300} = \frac{S_{BC}}{\sin 30}$$

$$\therefore S_{BC} = -5.7735 \text{ kN} = -5773.5 \text{ N}$$

(Negative sign means compression)

Ans: 18

Sol: By method of sections cut CI, EI and EF and prefer the right hand side, the FBD of which is shown below.



For equilibrium:

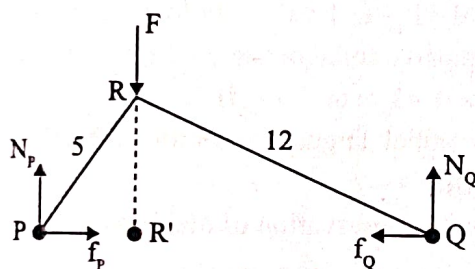
$$\Sigma M_E = 0 \text{ (Anticlockwise positive)}$$

$$T_{CI}(1.5) - 2(1.5) - 4(6) = 0$$

$$\Rightarrow T_{CI} = \frac{3 + 24}{1.5} = \frac{27}{1.5} = 18 \text{ kN}$$

66. Ans: 5.76

Sol:



By Pythagoras theorem: $PQ = 13 \text{ m}$

By Geometry:

$$5^2 - PR'^2 = 12^2 - QR'^2 \text{ and } [QR' = 13 - PR']$$

$$\Rightarrow PR' = \frac{25}{13} \text{ m}$$

$$\Rightarrow QR' = \frac{144}{13} \text{ m}$$

$$\Sigma M_P = 0$$

$$\Rightarrow F(PR') = N_Q(PQ)$$

$$\Rightarrow N_Q = \frac{25}{169} \cdot F$$

$$\Sigma F_y = 0$$

$$\Rightarrow N_P = F - \frac{25}{169} \cdot F = \frac{144}{169} F$$

$$\text{Now, } f_P = \mu_P \cdot N_P, \quad f_Q = \mu_Q \cdot N_Q$$

$$\Sigma F_x = 0$$

$$\Rightarrow \mu_P N_P = \mu_Q N_Q$$

$$\frac{\mu_Q}{\mu_P} = \frac{N_P}{N_Q} = \frac{144}{25} = 5.76$$

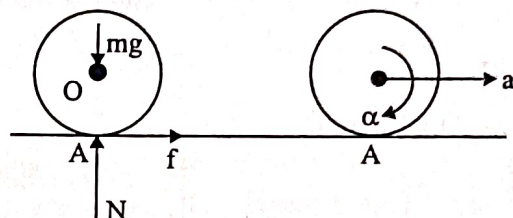
67. Ans: 0.25

Sol: Method I:

$$\text{Given: } (v_{\text{centre}})_{\text{initially}} = 0, \quad \omega_{\text{initial}} = 5 \text{ rad/s}$$

$$m = 1 \text{ kg}, \quad r = 0.15 \text{ m}$$

$$(v_{\text{centre}}) = ? \text{ [when rolling without slip starts]}$$



$\therefore$ Slip is backwards so friction will act in forward direction.

$$\Rightarrow f = ma$$

$$\mu mg = ma \Rightarrow a = \mu g$$

Let after time 't' the disc starts pure rolling.

Speed v of centre of mass after time 't'

$$v = u + at = \mu gt \quad \text{----- (i)}$$

Torque about centre of mass,

$$\tau = f \times r = \mu mgr$$

$$\tau = I \alpha$$

$$\mu mgr = \frac{mr^2}{2} \times \alpha$$

$$\alpha = \frac{2\mu g}{r} \text{ (Anticlockwise)}$$

Angular speed ω after time 't'

$$\omega = \omega_0 - \alpha t$$

$$\omega = 5 - \frac{2\mu g}{r} t \quad \text{----- (ii)}$$

For pure rolling,

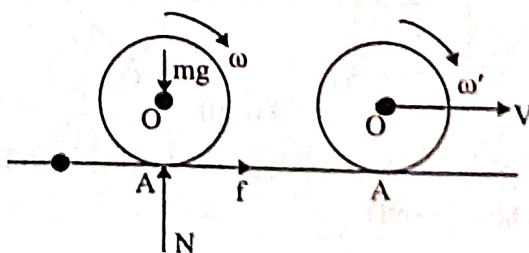
$$v = \omega r$$

$$\text{From (i) and (ii), } \mu gt = 5r - 2\mu gt$$

$$\Rightarrow t = \frac{5r}{3\mu g}$$

$$\therefore v = \mu g \times \frac{5r}{3\mu g} = \frac{5r}{3} = \frac{5}{3} \times 0.15 = 0.25 \text{ m/s}$$

Method II:



About point P there is no external torque.
[Because Torques of 'N' and 'mg' balances and friction force pass through point P, hence its torque is zero]

$$\therefore \left[\frac{d\vec{L}}{dt} = \vec{\tau}_{\text{ext}} \right]_{\text{about P}} = 0 \quad \left[\begin{array}{l} \text{where, } \vec{\tau} = \text{Torque} \\ \vec{L} = \text{Angular momentum} \end{array} \right]$$

$$\Rightarrow \vec{L}_i = \vec{L}_f$$

$$I_0 \omega = I_0 \omega' + mvr$$

$$\frac{mr^2}{2} \omega = \frac{mr^2}{2} \omega' + m\omega' r^2 \quad \left[\because v = \omega' r, \text{ because } \right. \\ \left. \text{pure rolling starts} \right]$$

$$\therefore \omega' = \frac{\omega}{3} = \frac{5}{3}$$

$$\Rightarrow v = \omega' \times r = \frac{5}{3} \times 0.15 = 0.25 \text{ m/s}$$

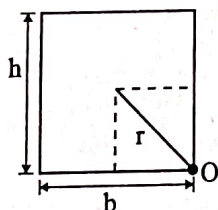
68. Ans: (a)

Sol: Mass moment of inertia of block about hinge 'O'

$$= \frac{M}{12} (h^2 + b^2) + Mr^2$$

where, $h = 0.4 \text{ m}$, $b = 0.3 \text{ m}$

$$r = \sqrt{0.15^2 + 0.2^2} = 0.25 \text{ m}$$



$$I_0 = \frac{1}{12} \times (0.4^2 + 0.3^2) + 1 \times 0.25^2 = 0.083 \text{ kg-m}^2$$

For block to topple about 'O' it should just reach the position such that centre of mass reaches from point (G) to point (G')

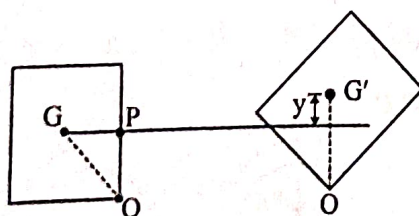


Fig. (i)

[Beyond this the torque of gravity will rotate the block itself]

$$\begin{aligned} \text{Now, } y(\text{increase in height of centre of mass}) &= OG' - OP \\ &= OG - OP \\ &= 0.25 - 0.20 = 0.05 \end{aligned}$$

So the block should have initial angular velocity just shift centre of mass at this position (i.e., G'). F.B.D of block is shown below.

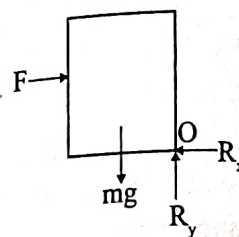


Fig. (ii)

$$\frac{d\vec{L}}{dt} = \vec{\tau}_{\text{ext}} \text{ (about O)}$$

[$\vec{L}$ is angular momentum, $\vec{\tau}_{\text{ext}}$ is external torque]

$$\therefore \tau_{\text{ext}} dt = d\vec{L}$$

$F\Delta t \times d = L_f - L_i$ [$\because$ Gravity force is small compared to impulsive force hence we neglect its torque]

$$I_f \times \omega = I_0 \times \omega \quad \text{--- (i)}$$

[ω is initial angular velocity of block just at impulse]

By using conservation of mechanical energy

$$\frac{1}{2} I_0 \omega^2 = mgy$$

$$\frac{1}{2} \times 0.083 \times \omega^2 = 1 \times 9.81 \times 0.05$$

$$\Rightarrow \omega = 3.44 \text{ rad/s}$$

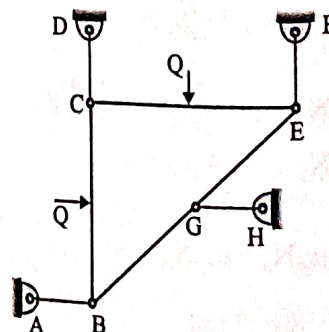
Putting in eq.(i)

$$I_f \times 0.3 = 0.083 \times 3.44$$

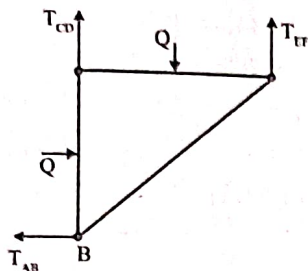
$$\Rightarrow I_f = 0.951 \text{ N-s}$$

69. Ans: (b, d)

Sol:



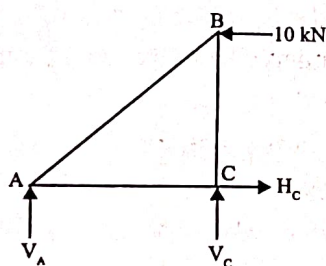
The force in member GH must be zero since GH is non collinear to BGE.



$$\begin{aligned}\Sigma F_x &= 0 \\ \Rightarrow T_{AB} &= Q \\ Q \text{ \& } T_{AB} \text{ create a clockwise couple which can be} \\ \text{overcome only by an anticlockwise couple for} \\ \text{which } T_{CD} &= 0, \\ \Rightarrow T_{EF} &= Q\end{aligned}$$

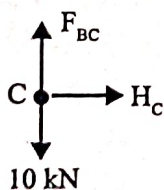
70. Ans: (b)

Sol: F.B.D of entire truss is shown below.



$$\begin{aligned}\text{For equilibrium, } \Sigma M_C &= V_A \times AC = 10 \times BC \\ \therefore V_A &= 10 \text{ kN} \\ \Sigma F_y &= 0 \therefore V_A + V_C = 0 \\ \therefore V_C &= -V_A = -10 \text{ kN}\end{aligned}$$

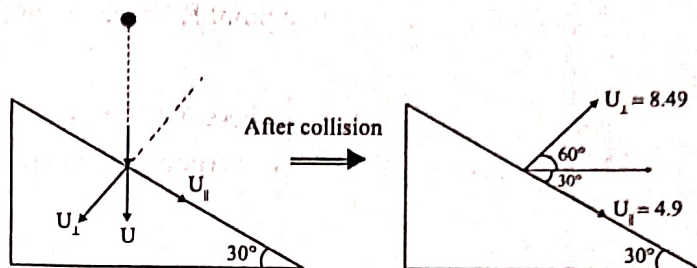
F.B.D of joint C



$$\begin{aligned}\Sigma F_y &= 0 \therefore F_{BC} = 10 \text{ kN} \\ \Rightarrow F_{BC} &= 10 \text{ kN (Tensile)}\end{aligned}$$

71. Ans: (c)

Sol:



Just before colliding rigid block velocity of spherical ball, $U = \sqrt{2 \times g \times h}$
 $= \sqrt{2 \times 9.8 \times 4.9} = 9.8 \text{ m/s}$

$U_{\parallel}$ = Component of velocity parallel to inclined plane after collision = Component of velocity parallel to inclined plane before collision

[$\because$ Force parallel to inclined plane is zero]

$$U_{\parallel} = U \cos 60^\circ = 9.8 \times \cos 60^\circ = 4.9 \text{ m/s}$$

$U_{\perp}$ = Component of velocity parallel to inclined plane after collision.

$$U_{\perp} = U \sin 60^\circ = 9.8 \times \sin 60^\circ = 8.49 \text{ m/s}$$

[$\because$ Collision is perfectly elastic ($e=1$) so velocity will just reverse its direction]

Thus resolving velocities ($U_{\parallel}$ & $U_{\perp}$) along $\hat{i}$ and $\hat{j}$ direction.

$$U_x = 4.9 \cos 30^\circ + 8.49 \cos 60^\circ = 8.48 \hat{i}$$

$$U_y = 8.49 \sin 60^\circ - 4.9 \sin 30^\circ = 4.9 \hat{j}$$

$$\text{Velocity vector after collision, } \vec{V} = 8.48 \hat{i} + 4.9 \hat{j}$$

Momentum vector after collision,

$$\begin{aligned}\vec{P} &= m\vec{V} = 2 \times (8.48 \hat{i} + 4.9 \hat{j}) \\ &= 17 \hat{i} + 9.8 \hat{j}\end{aligned}$$